# EXPERIMENT 4: LAN SETUP AND CONFIGURATION

## AIM

To set up and configure a Local Area Network (LAN) using a switch and Ethernet cables, verify connectivity between host machines, and share files across the network.

## REQUIREMENTS

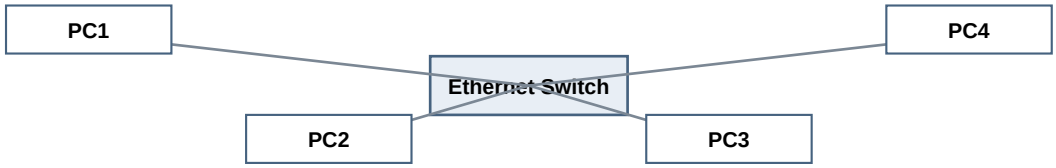
Four computers, one Ethernet switch, four Ethernet cables, and Windows network configuration tools.

## PROCEDURE

1. Connect four computers to the switch using Ethernet cables.
2. Assign each computer a unique IP address in the same network.
3. Verify connectivity between the computers using the `ping` command.
4. Share a folder with users on the same LAN and access it from another computer.

## LAN TOPOLOGY

**Star Topology Using an Ethernet Switch**




## IP CONFIGURATION

| Device | IP Address | Subnet Mask |
|---|---|---|
| PC1 | 10.1.1.1 | 255.0.0.0 |
| PC2 | 10.1.1.2 | 255.0.0.0 |
| PC3 | 10.1.1.3 | 255.0.0.0 |
| PC4 | 10.1.1.4 | 255.0.0.0 |
| Switch management IP* | 10.1.1.5 | 255.0.0.0 |

## CONNECTIVITY TEST

Command executed from PC1:

```
ping 10.1.1.2
```

| Sample Output | Meaning |
|---|---|
| Reply from 10.1.1.2: bytes=32 time<1ms TTL=128 | PC2 is reachable from PC1. |
| Packets: Sent = 4, Received = 4, Lost = 0 (0% loss) | Successful communication with no packet loss. |

## OBSERVATIONS

The computers connected to the switch communicated successfully when unique IP addresses from the same network were assigned. The ping test verified connectivity, and a shared folder could be accessed by other authorised computers on the LAN.

## OUTCOME AND CHALLENGES

Outcome: A four-computer LAN was configured, connectivity was verified using ping, and file sharing was enabled across the network.
Challenges: Incorrect IP addresses or subnet masks, loose or faulty Ethernet cables, disabled network adapters, and file-sharing permissions can prevent communication. Correct configuration and checking the cable links resolve these issues.

## RESULT

The LAN was set up using a switch and Ethernet cables. IP addresses were assigned, connectivity was tested successfully using ping, and file and folder sharing across the LAN was verified.